

RELATIONSHIP BETWEEN J-SPACE AND MODEL CHAIN OF THOUGHT

ABSTRACT

Gurnee et al. (2026) find that ablating the J-space (the model’s verbalizable workspace) destroys multi-step reasoning while sparing most capabilities, and that GSM8K with chain of thought (CoT) is far more robust to the ablation than direct answering, with written CoT externalizing the workspace. We test whether that substitution transfers to Qwen3-4B and how it behaves as problems get harder, finding a dissociation. The externalization mechanism is directly observable, where during CoT, the model holds each upcoming intermediate value in the J-space about 3 nats above its post-write loading and releases it once written (93 of 116 cases, $p < 10^{-10}$). However, the protective consequence does not transfer. No ablation strength in a 24-configuration sweep damages reasoning while preserving general coherence, the ablation damages one-hop recall more than two-hop reasoning, and after subtracting an operator-matched control, the J-specific damage is equal across direct and CoT modes. On Qwen3-4B, it appears that writing the steps down does not shield them.

1 HYPOTHESIS

All hypotheses were recorded before the runs that test them.

- H1. If CoT externalizes the workspace, then damaging the workspace should hurt silent answering more than written-out answering, and the gap should shrink as problems get harder because each written step still requires internal computation between tokens. Operationally, the test is valid only at an ablation strength that halves two-hop accuracy while leaving general function intact. The choice of halving is a pre-determined threshold. The original paper reports that its ablation drove multi-hop accuracy to near zero, so demanding a mere halving is relatively lenient; we are not requiring Qwen3-4B to match the paper’s full effect but we still wish to show an unmistakable one. We called this requirement the coherence gate: the ablated model’s next-token predictions on plain text must still match the clean model’s at least 85% of the time, with well-formed generations. Under those conditions, H1 predicts:
- (i) On GSM8K, the relative accuracy drop from J-ablation is at least twice as large for direct answering as for CoT answering, and a norm-matched random ablation shows no such gap.
 - (ii) The fraction of CoT accuracy that survives ablation shrinks faster with difficulty (MATH-500 levels 1–5, with AIME as an anchor) under J-ablation than under random ablation.

Alongside H1, we add two checks on whether the measurement itself could be trusted:

- H2. The paper’s ablation *sparcs* certain tokens, meaning it leaves a direction untouched whenever that token appears in the clean model’s top-10 next-token predictions at that position. The rule exists so that the ablation damages internal content rather than simply deleting the words the model is about to emit. But it has an asymmetric side effect because during a chain of thought, the digits the model is about to write are usually in the clean top-10 at exactly the moments that matter, so CoT gets its most critical tokens shielded automatically while direct answering does not. Sparing alone could therefore manufacture “CoT is more robust”.

Thus, H2 runs both configurations in conjunction, on the same problems with paired seeds: every primary condition follows the paper’s sparing rule, and a matching diagnostic run repeats the identical cells with sparing disabled. Neither configuration is interpreted on its own. A no-sparing run by itself is not a fair version of the paper’s experiment since without the exemption, the ablation can trivially delete the token the model was about to emit, and a sparing-on run by itself cannot rule out mechanical shielding.

The evidence is the comparison between the two: if the direct-vs-CoT gap appeared only with sparing on, this experiment could not distinguish workspace externalization from mechanical shielding, so H2 requires the gap measured with sparing on to survive at half its magnitude or more in the matching no-sparing run.

- H3. The paper itself found that numbers load weakly into the J-space, so if numeric tokens make up less than 10% of what the ablation actually removes during math, null results on math cannot count as evidence about the workspace.

Our operating-point search (Section 4) found that *no* ablation strength passes H1’s coherence gate on this model, so H1 is untestable as stated. Thus, we recorded two replacement hypotheses, again before their runs, designed to stay interpretable even when every available ablation damages general function along with reasoning.

- H4. Generic damage should hurt both answer modes about equally, so workspace externalization predicts *extra* protection specifically for CoT. Concretely, the relative drop under J-ablation is at least 1.5 times larger in direct mode than in CoT mode, and this gap is absent under a fair control that applies the identical ablation operator to randomly chosen tokens’ directions.

- H5. Two ablation-free measurements:

- (a) When a two-hop question demands an immediate answer, the model must resolve the hidden bridge entity (Carnival $\rightarrow$ Brazil) internally right away, so the bridge should load into the J-space harder than when the prompt invites step-by-step reasoning.
- (c) During CoT, each numeric intermediate should appear in the J-space *before* the model writes it out: the workspace holding a value, then releasing it to the page.

2 EXPERIMENT DESIGN

2.1 GATES BEFORE EXPERIMENTS

A *gate* is a go/no-go check whose pass threshold was frozen before any run: the main experiments proceed only through a gate that passes, and a failure is itself a reportable finding rather than a reason to loosen the threshold. Three gates precede the experiments.

Gate A. A workspace layer band exists and is identifiable on this model.

Gate B. Readout positive control. A *readout* is a passive measurement: the lens converts the model’s internal state at one layer and token position into a ranking over all vocabulary tokens, showing what the workspace is holding there without changing the model’s computation. The gate requires that on the paper’s own two-hop prompts, the hidden bridge entity appears among the top-10 tokens of that ranking for at least 12 of 20 items.

Gate C. An ablation operating point exists that is simultaneously coherent and effective, found by sweeping layer bands and ablation strength k .

2.2 THE ABLATION

At every token position and every band layer, we remove the k most strongly (positively) activated J-lens directions by exact span projection, sparing any token in the clean model’s top-10 predictions at that position (the exemption defined with H2 above). During generation this rule needs care, because an ablated model soon produces different text than a clean model would, and the paper does not specify which context the clean top-10 should be computed on. We resolve it with a paired clean forward pass at every decode step: an unablated copy of the model reads exactly the tokens the ablated model has produced so far and reports what it would predict next, and that prediction defines the spare set. Sparing against a clean model’s own separately generated text would instead protect tokens belonging to a sentence the ablated model is not writing.

2.3 CONTROLS

When accuracy falls under J-ablation, the drop could have several causes besides the loss of workspace content. Each control below isolates one alternative explanation, and each runs on the same problems as the main conditions.

1. **Random-token control.** Alternative explanation: removing that much of any activation is damaging, regardless of what it encodes. We therefore apply the identical removal procedure to the directions of randomly chosen vocabulary tokens instead of the most strongly activated ones. If the damage came from the surgery itself, this control would hurt the model too. If it comes from deleting content the model is actively using, the control should be nearly harmless.

2. **Renormalization check.** Alternative explanation: removing directions shrinks the activation vector, and later layers might misbehave because of the size change alone. We rerun the ablation but rescale each edited activation back to its original length afterward. Damage that persists is caused by the missing content, not the changed size.
3. **One-hop factual control.** Alternative explanation: the ablation damages knowledge retrieval in general rather than multi-step reasoning in particular. We add 30 single-step factual questions (for example, the chemical symbol for gold), scored identically to the 93 two-hop items. Damage to a reasoning workspace should hurt two-hop reasoning more than one-hop recall. If one-hop is hit as hard or harder, the damage is generic.
4. **The paper’s own capability criterion.** Alternative explanation: our coherence gate might be an unfairly strict measure of general function. We therefore also evaluate MMLU and an exact-copy task, the battery the original authors used to show their ablation spared general capabilities, so selectivity can be judged by their standard as well as ours.

2.4 MEASUREMENT

All generation cells use paired seeds (the seed depends on the problem and sample index, never the intervention condition), report truncation and format-break rates beside every accuracy, and are analyzed with a paired bootstrap clustered by problem.

2.5 MAIN EXPERIMENTS

2.5.1 EXPERIMENT 1 (ABLATION)

Tests H4. Every GSM8K problem is answered six ways: in two modes (a direct answer, or a written chain of thought) under each of three interventions (no ablation, J-ablation, and the random-token control). For each mode we compute the relative drop, meaning the fraction of clean accuracy that J-ablation removes in that mode. H4 claims the direct-mode drop is at least 1.5 times the CoT-mode drop, so the pre-registered test statistic is $\text{drop}_{\text{direct}} - 1.5 \text{ drop}_{\text{CoT}}$, where each term is that mode’s relative drop: a positive value means the asymmetry is at least as large as H4 requires. We bootstrap a confidence interval for this quantity, and H4 counts as supported only if the entire interval lies above zero.

2.5.2 EXPERIMENT 2 (READOUT)

Tests H5 and never modifies the model: it only reads. Every measurement is a lens readout, the passive ranking over vocabulary tokens defined at Gate B, which shows how strongly the workspace is holding each token at a given layer and position. For H5(a), each two-hop question is shown twice with different opening instructions, one requiring an immediate answer and one inviting step-by-step reasoning. The question tokens themselves are identical in both versions, so any difference in how strongly the bridge entity loads at the last five question tokens can only come from the instruction, not from the text being read. Loading is measured as the bridge token’s rank in the full-vocabulary lens readout. For H5(c), we let the model generate a chain of thought, find where each gold intermediate value first appears in it, and compare the lens log-probability of that value in the ten tokens just before it is written against the ten tokens just after. If the workspace holds a value and then releases it to the page, the before-window should score higher than the after-window.

3 EXPERIMENTAL DETAILS

3.1 MODEL AND SAMPLING

Qwen/Qwen3-4B in bf16 on one A100-80GB, sampled per the model card (never greedy). Direct-mode cells prefill the assistant turn with `\boxed{`, because without it the clean model ignores the answer-only instruction and truncates 83% of the time.

3.2 LENS FIT

The official `anthropics/jacobian-lens` estimator over 150 FineWeb prompts with the penultimate-layer target. Forensics on the fit found two heavy-tail outlier prompts carrying 47.5% of the summed per-prompt norm, and they were removed by exact checkpoint subtraction (final lens $n=148$).

3.3 DATASETS AND SCORING

Scoring extracts the final `\boxed{}` answer and checks it with `math-verify`.

- **GSM8K:** `openai/gsm8k`, test split.
- **MATH-500:** `HuggingFaceH4/MATH-500`.
- **AIME:** AIME 2024 (`Maxwell-Jia/AIME_2024`) plus AIME 2025 (`yentinglin/aime_2025`), pre-registered as thinking-mode-only and not reached (§5).

4 EXPERIMENTAL RESULTS

4.1 GATE OUTCOMES

- Gate A. Passes: a workspace band exists at layers **12–28** of 36, identified by readout autocorrelation and next-token-agreement profiles (the motor regime begins around L29), matching the published Qwen3.5-4B precedent. Kurtosis behaved anomalously and our occupancy metric was uninformative, and both are reported rather than hidden.
- Gate B. Fails: 5/20 against the 12/20 bar (the pre-registered logit-lens arbiter scored 6/20). The J-lens median rank was about 20 in a 151,936-token vocabulary, against about 53 for the logit lens, which is a real but diffuse signal. Ranks like 20 out of 151,936 are far above chance, so the measurement itself works. We concluded instead that the 12/20 bar, chosen before we knew how diffuse a 4B model’s readouts are, was set too high for a model of this size. We decided to continue to the next gate despite the failed threshold, and recorded that decision and its reasoning in the project log.
- Gate C. Fails: no viable operating point exists (Fig. 1). Across 24 configurations covering layer bands, k from 1 to 10, both selection semantics, and attention-sink handling, strong reasoning damage appears only at broken coherence (two-hop relative drop 0.80 at top-1 match around 0.45–0.56), and coherence recovers only where the effect vanishes (drop near 0 at top-1 0.73–0.83). The pre-registered viable region is never approached. Restoring the residual norm after ablation changes nothing (top-1 0.456 vs. 0.447), so the damage is content removal rather than norm shock.

4.2 WHAT THE ABLATION ACTUALLY DAMAGES

Gate C leaves a question open: the ablation clearly breaks something, so what is it actually removing? We examine this at the sweep’s strongest configuration, all band layers with $k=10$, hereafter the *strong operating point*, scoring all 93 two-hop items plus the 30-item one-hop control by teacher-forcing. The damage turns out to be generic but sharply J-specific (Table 3).

1. **Generic:** one-hop recall is damaged *more* than two-hop reasoning ($\Delta \log p$ of the answer’s first token +4.07 vs. +3.21 nats), and MMLU collapses from 0.76 to 0.45. By the paper’s own capability criterion, this ablation does not selectively impair reasoning.
2. **J-specific:** the operator-matched random-token control is nearly harmless ($\Delta \log p$ near 0 on both tasks, MMLU 0.65, top-1 match 0.75), so the damage comes from removing the specific directions the model is actively using, not from the sheer amount of activation removed: the control deletes a comparable amount and causes almost no harm.
3. **Validity checks:** Disabling sparing increases the measured effect slightly, in H2’s predicted direction. Consistent with H3, the deepest two-hop failures concentrate on abstract and numeric bridge entities, which is the paper’s own number-word caveat surfacing in our data.

4.3 EXPERIMENT 1: H4 IS REJECTED, CoT CONFERS NO J-SPECIFIC PROTECTION

GSM8K accuracies under each condition are given in Table 1. In parentheses, each ablated cell shows its change relative to that mode’s clean accuracy, where negative means accuracy lost.

Table 1: Experiment 1: GSM8K accuracy for each answer mode under each intervention. Brackets are 95% confidence intervals (Wilson); parentheses are the change relative to that mode’s clean accuracy. All cells use the same 75 problems with the same sampling randomness (direct mode originally ran 150; the mid-run reduction is recorded in the project log), and an answer is correct only if a parseable boxed answer matches. The control is statistically indistinguishable from clean (paired McNemar test, $p = 0.55$ direct, $p = 0.38$ CoT), so its small CoT-mode gain is sampling noise. The H4 test does not use these marginal intervals; it uses the tighter paired bootstrap over problems.

	Direct answer	Chain of thought
No ablation (clean)	0.307 [0.21, 0.42]	0.947 [0.87, 0.98]
J-space ablated	0.120 [0.06, 0.21] (−61%)	0.427 [0.32, 0.54] (−55%)
Random-token control	0.267 [0.18, 0.38] (−13%)	0.987 [0.93, 1.00] (+4%)

4.3.1 OUTCOME

H4 asked whether the direct-mode drop is at least 1.5 times the CoT-mode drop. From the table, the ratio is $61/55 = 1.11\times$. The pre-registered statistic, $\text{drop}_{\text{direct}} - 1.5 \text{drop}_{\text{CoT}} = -0.21$ with confidence interval $[-0.42, -0.02]$, lies entirely below zero, so H4 is rejected. Direct mode was originally run on 150 problems, of which these 75 are a subset. On the full 150 the numbers barely move (direct drop −64%, ratio 1.17 $\times$, statistic −0.18 with interval $[-0.36, -0.004]$), so the rejection is the same either way.

4.3.2 WHY THE CoT DROP IS SMALLER THAN IT LOOKS

Under ablation, 36% of CoT generations hit the token cap and 47% never produced a parseable answer, so part of the CoT drop is broken output rather than broken reasoning. The pre-registered format-robust rescoring (accepting the last number anywhere in the output) lifts the ablated CoT condition to 0.547, showing that about 12 points of its strict-scoring drop is generation collapse rather than wrong reasoning. After subtracting the fair control, the J-specific damage is essentially equal across modes (about 0.48 direct vs. 0.47 CoT). Writing the steps down does not shield them. Because CoT is fully robust to the control ablation, this is not a generic fragility of long generations.

4.4 EXPERIMENT 2: VALUES LOAD BEFORE THEY ARE WRITTEN, BUT NO HARDER WHEN AN IMMEDIATE ANSWER IS REQUIRED

4.4.1 OUTCOME

H5(a) is null: each question was shown twice, once under an instruction requiring an immediate answer and once under one inviting step-by-step reasoning. If the workspace were recruited harder when everything must be resolved internally, the hidden bridge entity should load more strongly in the immediate-answer version. It does not: over 84 items whose question tokens are identical in both versions, bridge loading is indistinguishable (Table 2). How strongly a value loads is set by the question itself, not by what the model is about to be asked to do with it.

H5(c) is confirmed (Fig. 2): as the model approaches the moment of writing an intermediate value, the value is loaded in the J-space, and once it is on the page the loading drops away, as if the written text takes over the job of remembering it. Across 116 located intermediates, loading is higher in the ten tokens before writing than in the ten after for 93 of 116 values (mean lens log-probability −5.6 before vs. −8.4 after, about a 3-nat release; sign test $p < 10^{-10}$). The effect persists from GSM8K through MATH level 5 with no monotone difficulty trend.

Table 2: Experiment 2, H5(a): bridge-entity loading at the last five question tokens is the same whether the prompt requires an immediate answer or invites step-by-step reasoning ($n = 84$ items). Rank is the bridge token’s position in the full-vocabulary lens readout (lower means loaded more strongly). Pairing the two versions of each item, the bridge loads higher (by lens log-probability) under the immediate-answer instruction in 41 items and under the step-by-step instruction in 43 (exact sign test $p = 0.91$).

	Immediate answer	Step-by-step
Median bridge rank	6.0	6.0
Rank interquartile range	[2, 69]	[2, 71]
Bridge in lens top-10	60%	60%
Mean lens log-probability	−5.27	−5.22

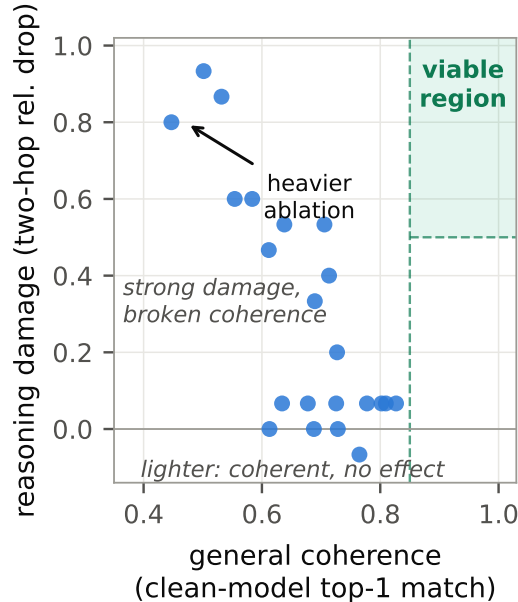


Figure 1: We ran the same ablation at 24 different strengths, varying how many J-lens directions are removed and at how many layers. Each dot is one such run, placed by how intact the model remains overall (x axis) and how much two-hop reasoning it loses (y axis). Heavier runs (upper left) break reasoning only by also breaking the model in general, lighter runs (lower right) leave the model intact but change nothing, and no run reaches the viable region at top right, where reasoning would be damaged while the model stays intact. Both of the region’s boundaries were fixed before any run: coherence at least 0.85 (the coherence gate from H1) and two-hop relative drop at least 0.50 (the halving requirement from H1), so the region marks where a run would have to land for the main experiment to be valid, not a line fit to these points. Two runs re-measured with the final corrected implementation lie on the same trade-off.

Table 3: What each intervention breaks. Each row is one intervention, measured on the same items with the same four instruments; the J-space and random-token rows use the strong operating point of Section 4.2, and the thin-window row is a gentler configuration from the same sweep, shown for contrast. Damage is the drop in teacher-forced log-probability of the correct answer’s first token, in nats, relative to the clean model (larger is worse, near zero is harmless). Coherence is the fraction of next-token predictions on plain text that still match the clean model. Dashes mark cells that were not measured: MMLU was run only where a conclusion depends on it, and the dashed rows are diagnostics whose other columns already settle their question.

Condition	Two-hop damage	One-hop damage	Coherence	MMLU-100
No ablation (clean)	0	0	1.00	0.76
J-space, $k=10$	+3.21	+4.07	0.45	0.45
J-space + renormalization	+3.19	+3.92	0.46	–
Thin window, $k=10$	−0.18	−0.04	0.73	–
Random tokens (mean of 3 seeds)	−0.12	+0.17	0.76	0.65

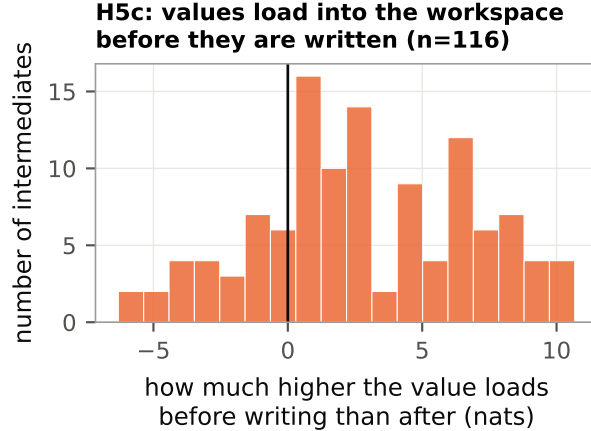


Figure 2: Experiment 2, H5(c): for each of 116 intermediate values in generated chains of thought, how much more strongly the value loads in the workspace just before it is written than just after.

Interpretation of Figure 2: The clean model writes a chain of thought, and we locate the token where each required intermediate value (say, a subtotal of 36) first appears. Each histogram entry is one such value, placed by the difference between how strongly the lens reads it in the ten tokens just before it is written and in the ten tokens just after. A value to the right of the zero line was held in the workspace on the approach to being written and released once it was on the page, and that is where the distribution sits: 93 of 116 values. This is the externalization mechanism operating on Qwen3-4B, caught with no intervention at all, and it is what makes Experiment 1’s result a dissociation, showing that the mechanism is present yet it protects nothing.

5 ANALYSIS OF RESULTS

5.1 FINDING 1: THE SELECTIVE ABLATION RESULT DOES NOT TRANSFER TO QWEN3-4B

No strength of adaptive top- k J-ablation on Qwen3-4B impairs reasoning while sparing general function, and once a one-hop control was in place, the impairment turned out never to have been reasoning-specific in the first place. Our reading is that at 4B scale, the most strongly activated verbalizable directions carry the model’s generic knowledge-driven prediction, so ablating the workspace is indistinguishable from ablating knowledge. We state this as a search-outcome claim about this operator family on this model. The conclusion also does not depend on the exact 0.85 coherence bar: no strong-effect run exceeds top-1 match 0.56, coherence recovers only past 0.73 where the effect is already gone, and the strong operating point independently fails the paper’s own capability criterion (MMLU 0.45 vs. 0.76 clean). The phrasing follows a four-way adversarial review, whose two confirmed instrument confounds (anti-correlated coherence metrics, resolved by running the paper’s own battery, and an unfair random control) were fixed and re-measured before concluding.

5.2 FINDING 2: THE EXTERNALIZATION MECHANISM IS REAL, BUT IT DOES NOT PROTECT

The externalization *mechanism* is real and directly observable without any ablation. During CoT the model holds each upcoming intermediate about 3 nats above its post-write loading and then releases it to the page (H5c), which is exactly the paper’s picture, verified at 4B. But the *protective* consequence claimed by the paper does not appear. The ablation damages CoT-mode and direct-mode reasoning equally once generation collapse and control damage are accounted for (H4 rejected), and the workspace loads no harder when the prompt requires an immediate answer (H5a null). Both fit the mechanism story: if the strongly activated J-directions carry generic knowledge processing rather than a separable working memory, then writing intermediates down cannot rescue what the ablation actually breaks, and there is no extra holding capacity for the model to recruit when it must answer immediately.

5.3 SCOPE AND NATURAL EXTENSIONS

These findings are established on a single model with a single lens fit, and every measurement, deviation, and instrument change is logged in the repository so the boundaries of the claims are explicit. Two scoping choices matter most

when reading them. First, the difficulty axis: the pre-registered MATH and AIME survival curves required an ablation operating point that Gate C showed does not exist on this model, so the difficulty question is answered here by the readout experiment, where the holding effect persists unchanged from GSM8K through MATH level 5. Second, the conclusions concern the adaptive top- k ablation family. We chose this scope because it is the procedure the original paper used, and whether that procedure transfers is the question this study set out to answer. Each boundary doubles as a concrete, falsifiable prediction for follow-up work. Fixed bridge-entity ablations (the intervention the original paper demonstrated causally) test whether targeted removal succeeds where content-agnostic removal cannot. The differential design is ready to run on the MATH and AIME gradient on a model competent enough for those problems. And replication above 4B, where lens signal concentrates, tests our central interpretation directly: if workspace and knowledge directions disentangle with scale, the paper’s selectivity result should reappear there.

REFERENCES

Wes Gurnee, Nicholas Sofroniew, et al. Verbalizable representations form a global workspace in language models. *Transformer Circuits Thread*, 2026. URL <https://transformer-circuits.pub/2026/workspace/index.html>.